Yongda Zhu, Ph.D.
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RESEARCH INTERESTS

I study the evolving intergalactic medium (IGM) and its connection to galaxies and quasars.

Keywords: Intergalactic medium; galaxies; quasars; reionization; AI×JWST/Roman

POSITIONS

- **JASPER Postdoctoral Scholar**, University of Arizona 2025–Present
  Mentors: Prof. Eiichi Egami & Prof. Xiaohui Fan
  Leading NIRCam/WFSS analysis for SAPPHIRES and studying gas–galaxy interactions across scales with JWST and machine learning.
- **Postdoctoral Research Associate**, NIRCam contract 2024–Present
  Mentors: Prof. Marcia Rieke & Prof. George Rieke
  Member of JWST NIRCam Science Team and MIRI U.S. Team.
  Led a systematic search for galaxies with extended line emission and candidate outflows using deep NIRCam medium-band imaging and machine learning.
  Leading the SMILES (JWST-GTO-1207) NIRSpec public data release and multi-band NIRCam photometric analysis of galaxies.
  Leading reduction and analysis of JWST/NIRSpec MSA spectroscopy of cosmic-noon galaxies and obscured AGN.
- **Graduate Student Researcher**, UC Riverside 2019–2024
  Mentor: Prof. George Becker
  Led Keck and ALMA observations of the IGM via high-z quasars; modeled the Lyα and Lyβ forest using cosmological simulations.
  Provided observational evidence for a late and spatially inhomogeneous end to reionization.

EDUCATION

Mar. **2024** **Ph.D. in Physics**; University of California, Riverside
Constraining the IGM during the Later Stages of Reionization using QSO Spectra
Advisor: Prof. George Becker
Dec. 2019 M.Sc. in Physics; University of California, Riverside
Jun. 2018 B.Sc. in Astronomy; University of Science and Technology of China
Testing Gravity Theories on Galactic Scales. Advisor: Prof. Xiao-Bo Dong

SUCCESSFUL PROPOSALS

Principal Investigator / Co-Principal Investigator Programs

- *Roman 10995 – *Establishing the Post-Reionization Lya Transmission-Environment Baseline*
- *Roman 19049 – *From Environment to Assembly: Roman Spatially Resolved Stellar Mass and SFH Maps*
- *Roman 19054 – *Do Filaments Influence Galaxies Beyond Local Environment? A Roman Census of Galaxy-Filament Connectivity* (as co-PI)
- *Roman 19029 – *Linking Galaxies to Their Neutral Gas Reservoir: A Roman–Rubin Synergy Survey of DLA Host Galaxies at Cosmic Noon* (as co-PI)
- *MMT-6.5m/Binospec IFU – 2026A: *IFU Observations of a $z = 5.63$ Quasar-Companion System: Lyα Escape and UV Counterpart*
- *MMT-6.5m/Binospec IFU – 2025B: *Spatially Resolved Lyα Spectroscopy of a $z = 5.5$ Quasar with Strong Cold Outflows*
- *ALMA – Cycle 11: *Galaxy Over/Underdensities Around IGM Transmission at $z = 5.7$: A Robust Constraint on Reionization*
- *MMT-6.5m/Binospec IFU – 2024B: *Ionization and Enrichment in the Reionization Epoch: A Pilot Study with Binospec IFU*
- *ALMA – Cycle 9: *The Mean Free Path of Ionizing Photons at $z = 5.6$: A Robust Constraint on Reionization*

As a student leading the observations: † UC students could not serve as Keck PIs

- JWST 4092 – *How Does Reionization End? A Search for [O III] Emitters in the Most Transparent Regions of the IGM Near Redshift Six* (PI: Becker)
  Zhu, et al. 2025, arXiv:2510.09568
- †Keck/ESI [2021B_U036, 2022A_U035, 2022B_U042, 2023B_U049, 2024A_U281]: *The Mean Free Path at $z = 5.6$: Insights into Ultra-Late Reionization*
- Keck/ESI [2019A_U014, 2020A_U121, 2021A_U039]: *Giant Lyα Troughs at $z < 6$: A Signature of Very Late Reionization?*
  Zhu, et al. 2021, ApJ, 923, 223; **Zhu**, et al. 2022, ApJ, 932, 76
- Keck/LRIS [2019A_U147, 2019B_U147]: *The Mean Free Path at $z = 5$: A Key Constraint on Reionization Models*
  Becker et al. 2021; **Zhu**, et al. 2023, ApJ, 955, 115
- Other allocations include: Keck/ESI [2021A, 2021B], Keck/DEIMOS [2020A], Keck/LRIS [2023B], Subaru/HSC [2020B, 2021A, 2021B, 2023B]

Selected Major Co-Investigator Programs:

- JWST 12583 – The Dragon Survey: A Direct Probe of the Early Stellar Luminosity Function and Dark Matter through Multi-cycle JWST Multi-cadence Microlensing at $z = 0.73$

- JWST 12577 – JWST Multi-Cycle Deep Transient Survey in GOODS-S
- JWST 10790 – LAPIS: Large Area Parallel Imaging Survey
- JWST 10311 – Slicing the Saguaro: Dissecting a Little Red Dot Descendant at Cosmic Noon with NIRSpec/IFU
- JWST 8544 – Rest-frame Optical Nebular Emission Lines at Cosmic Dawn: MIRI/LRS Follow-up for JADES-GS-z14-0
- JWST 8060 – JWST Multi-Cycle Deep Transient Survey in GOODS-S
- JWST 8018 – DIVER: Deep Insights into UV Spectroscopy at the Epoch of Reionization
- JWST 7935 – Efficient Measurement of the Emergence Rate of AGN in Legacy Deep Fields
- JWST 7436 – The Last Neutral Islands at the End of Reionization? Characterizing the Nature of the Longest Dark Gaps in IGM Transmission at $z \sim 5.3$
- JWST 7390 – Probing Pair-Instability Supernovae through the Triply-lensed MACS0647-JD at $z = 10.17$
- JWST 7345 – The Dragon Survey: A Direct Probe of the Early Stellar Luminosity Function and Dark Matter through Multi-cycle JWST Multi-cadence Microlensing at $z = 0.73$
- JWST 7336 – Commission a new $R \sim 2500$ "IFU" on JWST: Calibrating second-order spectra of NIRCam/WFSS through the Hubble Ultra Deep Field
- JWST 7335 – Forever Blowing Bubbles: What Powers a 24-kpc Ionized Gas Nebula Around a Normal Galaxy at $z = 6$?

PROFESSIONAL SERVICE

- **Grant Reviewer**: NSF Astronomy and Astrophysics Research Grants Review Panel 2026
- **Peer Reviewer**: *The Astrophysical Journal*, *ApJ Letters*, *JCAP*, *Open Research Europe*; ALMA and Gemini proposals
- **JWST Collaboration Leadership**: *SAPPHIRES* (Data Lead); *SMILES* (NIRSpec Lead); *COSMOS-3D* (LSS Working Group Co-chair)
- **Other Major Collaborations**: JWST/JADES, JWST/DIVER, ESO VLT XQR-30, Roman Science Collaboration

SELECTED AWARDS

- Roman Space Telescope Cycle 1 Programs, PI/co-PI
- NSF / NRAO Student Observing Support Award ($40,000) 2023
- UCR Dissertation Year Fellowship ($7,200) 2022

- Benjamin C. Shen Memorial Award for Outstanding First-Year Graduate Researcher, UCR 2019
- Dean's Distinguished Fellowship, UC Riverside 2018
- Undergraduate research awards and scholarships, USTC 2015–2018
  Includes CAS Undergraduate Research Award, NSFC Talented Students in Basic Sciences Program, CAS Innovation Training Program, and Outstanding Student Scholarships.

SELECTED INVITED TALKS & CONFERENCE PRESENTATIONS

* invited

- * Star-forming Galaxies in the Reionization Epoch, Ishigaki, Japan March 2026
- Galaxy Origins in the JWST Era, Toledo May 2025
- JWST NIRCam Science Meeting, Biosphere 2 March 2025
- * NOIRLab Joint Colloquium, **University of Arizona** February 2025
- * Cosmology Seminar, **Arizona State University** February 2025
- Lyman-alpha Forest Workshop, **Ohio State University** October 2024
- JWST MIRI Science Meeting, Biosphere 2 October 2024
- The First Gigayear(s), Hilo, Hawai'i September 2024
- JADES Collaboration Meeting, **University of Copenhagen** April 2024
- * Galaxy Group Talk, **University of Arizona** March 2024
- * Galaxy Seminar, **University of Michigan** November 2023
- Galaxy Formation and Evolution in Southern California September 2023
- * Kashiwa-Mitaka Meeting (KMM) Seminar, **University of Tokyo** August 2023
- * Lightning Talk, First Light Conference, **MIT** June 2023
- Reionisation in the Summer Conference, **MPIA**, Heidelberg June 2023
- * Galaxy Formation Group, **Northwestern University (CIERA)** December 2022
- * FLASH Seminar, **UC Santa Cruz** November 2022
- * Astronomy Lunch Talk, **UC Los Angeles** October 2022
- * Astro Lunch Talk, **UC Santa Barbara** September 2022
- Reionization on a Blackboard Workshop, **CCA** September 2022
- * Special arXiv Coffee, **UC Davis** May 2022
- * Physics & Astronomy Seminar, **UC Riverside** April 2022
- * High-z Group Talk, **Tsinghua University** April 2022

- Reionization and Cosmic Dawn, **UC Berkeley** March 2022
- European Astronomical Society Annual Meeting (EAS 2021) July 2021
- Summer All Zoom Epoch of Reionization Conference June 2021
- * EURECA Seminar, **University of Arizona** February 2021

MENTORING & TEACHING

- **Teaching Assistant**, UC Riverside 2018–2019, 2023
  Taught intensive PhD summer review course (Comprehensive Exam prep).
  Topics: Classical Mechanics, E&M, Thermo & Stat Mech
  Led undergraduate general physics labs.
- **Teaching Assistant**, USTC 2016
  Taught introductory programming in `C/C++`, including data structures and algorithms.
- **Graduate Student Research Mentoring** 2025–
  Ms. Zheng Ma (University of Arizona)
- **Undergraduate Students (Research Mentor)**
  - Mr. Suprabhas Narisetty (University of Arizona) 2025–
    Project: NIRSpec observations of cosmic-noon galaxies
  - Ms. G. Hernandez (UC Riverside) 2021
    Project: Evolution of IGM effective optical depth
- **Graduate Peer Mentoring**
  - Mr. Seyedazim Hashemi (PhD student, UC Riverside)
    Project: Lyman-alpha visibility during reionization
- **UCR International Students & Scholars Office** 2021–2023
  Mentored 14 international graduate students (career and academic guidance)
- **UCR Graduate Student Mentorship Program (GSMP)** 2020
  Mentored two junior PhD students across STEM fields:
  Dr. N. Ahvazi (Dark matter & galaxies); Dr. Q. Wu (2D materials)

PUBLIC OUTREACH & SERVICE

- Invited public talk at Arizona Senior Academy April 2026
- Session Chair, AAS 247 January 2026
- Chambliss Judge for student poster competition, AAS 247 January 2026
- Instructor, *Galactic Adventures*, Flandrau Science Center & Planetarium June 2025
- Stargazing outreach event, Home Gardens Library (Corona, CA) October 2023

- Co-organizer, UCR Physics & Astronomy Student Seminar 2022–2023
- Instructor, UCR Camp Highlander Summer 2022
- Designed and taught outreach courses for K–12 students 2022
  - *Multiwavelength Universe*, *Gravity Simulator*
- Virtual Stargazing (UCR & Riverside Astronomical Society): monthly live public outreach on YouTube 2020–2021
- Science fair judge, Riverside County Science and Engineering Fair 2021–2023
- UCR Astronomy Public Outreach: Mercury Transit November 2019

OPEN SCIENCE & COMMUNITY RESOURCES

- **SMILES NIRSpec HLSP (JWST-GTO-1207)**: Lead developer of the public high-level data release including 1D/2D spectra and derived galaxy catalogs.
  GitHub: github.com/ydzhuastro/smiles_dr2; MAST HLSP: doi:10.17909/et3f-zd57
- **Keck/ESI Rest-UV Spectra for Bright Quasars**: Reduced and calibrated spectra for $m_z < 21$ quasars at $z \sim 5.5$, publicly available on Zenodo.
  doi.org/10.5281/zenodo.18677315.
- **Tutorial: Galaxy Morphology $\times$ Machine Learning** February 2025
  Introducing morphology classification and neural network tools for students.
  EURECA Tutorial PDF

SELECTED MEDIA COVERAGE

- "Astronomers Discover a Unique Quasi-Stellar Object-Dusty Star-Forming Galaxy System," **Phys.org**
- "The End of the Cosmic Dawn: Settling a Two-Decade Debate," **SciTechDaily**
- "Can You Explain These Long, Dark Gaps in Your Cosmological Resume?" **AAS Nova**
- "Intergalactic Impacts of Quasars During the Epoch of Reionization" **AAS Nova**
- "Cosmic predators: How supermassive black holes slow star growth in nearby galaxies" **University of Arizona News**

YONGDA ZHU - PUBLICATION LIST

ORCID: 0000-0003-3307-7525
ADS link: https://ui.adsabs.harvard.edu/search/q=orcid%3A0000-0003-3307-7525
Google Scholar: https://scholar.google.com/citations?user=wDrSZWYAAAAJ

Citation Metrics (as of June 2026):
Total citations: 3500+ | h-index: 35 | First-author citations: 400+ | First-author papers: 14

First-author:

14. **Zhu, Y.**, Ji, Z., Becker, G. D., Ding, J., Egami, E., Fan, X., Jin, X., Liu, W., Lyu, J., Ma, Z., Narisetty, S., Rieke, G. H., Wu, Y., Yue, M., Zhang, J., and Rieke, M. J., **2026**. *Early metal-enriched baryon cycling before the midpoint of cosmic reionization*, arXiv e-prints, arXiv:2604.21218. Nature Astronomy, accepted in principle

13. **Zhu, Y.**, Becker, G. D., D'Aloisio, A., Endsley, R., Gangolli, N., Cain, C., Mason, C. A., Hashemi, S., and Hong, H., **2026**. *Galaxy Underdensities Host the Clearest Intergalactic Medium Lyα Transmission and Indicate Anisotropic Reionization*, The Astrophysical Journal, 1002, 93

12. **Zhu, Y.**, Rieke, M. J., Ho, L. C., Sun, Y., Rieke, G. H., Yuan, F., Bakx, T. J. L. C., Becker, G. D., Yang, J., Bañados, E., Bischetti, M., Cain, C., Fan, X., Fudamoto, Y., Hashemi, S., Ikeda, R., Ji, Z., Jin, X., Liu, W., Liu, Y., Lyu, J., Ma, H.-X., Takeuchi, T. T., Umehata, H., Wang, F., and Tee, W. L., **2026**. *A Potential Link between Nuclear Winds and Cold Gas Outflows on Kiloparsec Scales in Reionization-era Quasars*, The Astrophysical Journal, 1000, 312.

11. **Zhu, Y.**, Rieke, M. J., Ji, Z., Bunker, A. J., Carreira, C., Danhaive, A. L., Duan, Q., Egami, E., Eisenstein, D. J., Hainline, K., Johnson, B. D., Ma, Z., Puskás, D., Rieke, G. H., Rinaldi, P., Robertson, B., Tacchella, S., Übler, H., Villanueva, N. C., Williams, C. C., Willmer, C. N. A., Wu, Z., and Zhang, J., **2026**. *Clumps in High-redshift Galaxies: Mass Scaling and Radial Trends from JADES*, The Astrophysical Journal, 1000, 303.

10. **Zhu, Y.**, Bonaventura, N., Sun, Y., Rieke, G. H., Alberts, S., Lyu, J., Shivaei, I., Morrison, J., Ji, Z., Egami, E., Helton, J. M., Rieke, M. J., Rinaldi, P., Sun, F., and Willmer, C. N. A., **2026**. *SMILES Data Release. II. Probing Galaxy Evolution during Cosmic Noon and Beyond with NIRSpec Medium-resolution Spectra*, The Astrophysical Journal, 997, 301.

9. **Zhu, Y.**, Egami, E., Fan, X., Sun, F., Becker, G. D., Cain, C., Chen, H., Eilers, A.-C., Fudamoto, Y., Helton, J. M., Jin, X., Pudoka, M., Bunker, A. J., Cai, Z., Champagne, J. B., Ji, Z., Lin, X., Liu, W., Ma, H.-X., Ma, Z., Maiolino, R., Rieke, G. H., Rieke, M. J., Rinaldi, P., Sun, Y., Tee, W. L., Wang, F., Yang, J., Yue, M., and Zhang, J., **2025**. *Quasar Radiative Feedback May Suppress Galaxy Growth on Intergalactic Scales at z = 6.3*, The Astrophysical Journal, 995, L5.

8. **Zhu, Y.**, Rieke, M. J., Ji, Z., Simmonds, C., Sun, F., Sun, Y., Alberts, S., Bhatawdekar, R., Bunker, A. J., Cargile, P. A., Carniani, S., de Graaff, A., Hainline, K., Helton, J. M., Jones, G. C., Lyu, J., Rieke, G. H., Rinaldi, P., Robertson, B., Scholtz, J., Übler, H., Williams, C. C., and Willmer, C. N. A., **2025**. *A Systematic Search for Galaxies with Extended Emission Lines and Potential Outflows in JADES Medium-band Images*, The Astrophysical Journal, 986, 162.

7. **Zhu, Y.**, Alberts, S., Lyu, J., Morrison, J., Rieke, G. H., Sun, Y., Helton, J. M., Ji, Z., Bhatawdekar, R., Bonaventura, N., Bunker, A. J., Lin, X., Rieke, M. J., Rinaldi, P., Shivaei, I., Willmer, C. N. A., and Zhang, J., **2025**. *SMILES: Potentially Higher Ionizing Photon Production Efficiency in Overdense Regions*, The Astrophysical Journal, 986, 18.

6. **Zhu, Y.**, Bakx, T. J. L. C., Ikeda, R., Umehata, H., Becker, G. D., Cain, C., Champagne,

J. B., Fan, X., Fudamoto, Y., Jin, X., Ma, H.-X., Sun, Y., Takeuchi, T. T., and Tee, W. L., **2024**. *Discovery of a Unique Close Quasar–DSFG Pair Linked by a [C II] Bridge at z = 5.63*, Research Notes of the American Astronomical Society, 8, 284.

5. **Zhu, Y.**, Becker, G. D., Bosman, S. E. I., Cain, C., Keating, L. C., Nasir, F., D'Odorico, V., Bañados, E., Bian, F., Bischetti, M., Bolton, J. S., Chen, H., D'Aloisio, A., Davies, F. B., Davies, R. L., Eilers, A.-C., Fan, X., Gaikwad, P., Greig, B., Haehnelt, M. G., Kulkarni, G., Lai, S., Puchwein, E., Qin, Y., Ryan-Weber, E. V., Satyavolu, S., Spina, B., Walter, F., Wang, F., Wolfson, M., and Yang, J., **2024**. *Damping wing-like features in the stacked Ly α forest: Potential neutral hydrogen islands at $z < 6$*, Monthly Notices of the Royal Astronomical Society, 533, L49.

4. **Zhu, Y.**, Becker, G. D., Christenson, H. M., D'Aloisio, A., Bosman, S. E. I., Bakx, T., D'Odorico, V., Bischetti, M., Cain, C., Davies, F. B., Davies, R. L., Eilers, A.-C., Fan, X., Gaikwad, P., Haehnelt, M. G., Keating, L. C., Kulkarni, G., Lai, S., Ma, H.-X., Mesinger, A., Qin, Y., Satyavolu, S., Takeuchi, T. T., Umehata, H., and Yang, J., **2023**. *Probing Ultralate Reionization: Direct Measurements of the Mean Free Path over $5 < z < 6$*, The Astrophysical Journal, 955, 115.

3. **Zhu, Y.**, Ma, H.-X., Dong, X.-B., Huang, Y., Mistele, T., Peng, B., Long, Q., Wang, T., Chang, L., and Jin, X., **2023**. *How close dark matter haloes and MOND are to each other: three-dimensional tests based on Gaia DR2*, Monthly Notices of the Royal Astronomical Society, 519, 4479.

2. **Zhu, Y.**, Becker, G. D., Bosman, S. E. I., Keating, L. C., D'Odorico, V., Davies, R. L., Christenson, H. M., Bañados, E., Bian, F., Bischetti, M., Chen, H., Davies, F. B., Eilers, A.-C., Fan, X., Gaikwad, P., Greig, B., Haehnelt, M. G., Kulkarni, G., Lai, S., Pallottini, A., Qin, Y., Ryan-Weber, E. V., Walter, F., Wang, F., and Yang, J., **2022**. *Long Dark Gaps in the Lyβ Forest at $z < 6$: Evidence of Ultra-late Reionization from XQR-30 Spectra*, The Astrophysical Journal, 932, 76.

1. **Zhu, Y.**, Becker, G. D., Bosman, S. E. I., Keating, L. C., Christenson, H. M., Bañados, E., Bian, F., Davies, F. B., D'Odorico, V., Eilers, A.-C., Fan, X., Haehnelt, M. G., Kulkarni, G., Pallottini, A., Qin, Y., Wang, F., and Yang, J., **2021**. *Chasing the Tail of Cosmic Reionization with Dark Gap Statistics in the Lyα Forest over $5 < z < 6$*, The Astrophysical Journal, 923, 223.

Student-led papers:

2. Ma, Z. (UofA grad student), Egami, E., **Zhu, Y.**, Sun, F., Lyu, J., Zhang, J., Willmer, C. N. A., Bunker, A. J., Carniani, S., Curtis-Lake, E., Hausen, R., Ji, X., Ji, Z., Juodžbalis, I., Maiolino, R., Rieke, G. H., Rinaldi, P., Sun, Y., Tacchella, S., Übler, H., and Williams, C. C., **2026**. *Undermassive Hosts of $z = 4 - 6$ AGN from JWST/NIRCam Image Decomposition with CONGRESS, FRESCO, and JADES*, arXiv e-prints, arXiv:2601.15962. Submitted to ApJ.

1. Narisetty, S. (UofA undergraduate), **Zhu, Y.**, and Egami, E., **2025**. *Strong Balmer Breaks Mark Gradually Quenching Galaxies Below the Star-forming Main Sequence at $1 < z < 4$*

with JWST/NIRSpec, for ApJ submission. DOI: 10.5281/zenodo.17772416

Co-author:

83. Sun, Y., Ji, Z., D'Eugenio, F., **Zhu, Y.**, Rieke, G. H., Baker, W. M., Bunker, A. J., Carniani, S., Helton, J. M., Perna, M., Pérez-González, P. G., Rinaldi, P., Übler, H., and Willmer, C. N. A., **2026**. *A Census of Na D-traced neutral ISM and outflows at* $0.6 < z < 4$, arXiv e-prints, arXiv:2604.18522.

82. Lebowitz, S., Hainline, K. N., Juneau, S., Williams, C. C., Panda, S., Lyu, J., Maseda, M. V., Tacchella, S., **Zhu, Y.**, and Aguayo, J. L., **2026**. *Spatially Resolved AGN Ionization and Star Formation at Cosmic Noon with JWST/JEMS*, arXiv e-prints, arXiv:2604.01271.

81. Danhaive, A. L., Tacchella, S., McClymont, W., Robertson, B., Carniani, S., Carreira, C., Egami, E., Bunker, A. J., Curtis-Lake, E., Eisenstein, D. J., Ji, Z., Johnson, B. D., Rieke, M., Villanueva, N. C., Willmer, C. N. A., Willott, C., Wu, Z., and **Zhu, Y.**, **2026**. *Beyond the stars: Linking Hα sizes, kinematics, and star formation in galaxies at* $z \approx 4-6$ *with JWST grism surveys and GEKO*, Monthly Notices of the Royal Astronomical Society, 547, stag437.

80. Danhaive, A. L., Tacchella, S., Bunker, A. J., Curtis-Lake, E., de Graaff, A., D'Eugenio, F., Duan, Q., Egami, E., Eisenstein, D. J., Johnson, B. D., Maiolino, R., McClymont, W., Rieke, M., Robertson, B., Sun, F., Willmer, C. N. A., Wu, Z., and **Zhu, Y.**, **2026**. *The dark side of early galaxies: GEKO uncovers dark-matter fractions at* $z \sim 4-6$, Monthly Notices of the Royal Astronomical Society, 546, stag119.

79. Rowlands, S., Davies, R. L., Ryan-Weber, E., Keating, L. C., Sebastian, A. M., Becker, G. D., Bischetti, M., Bosman, S. E. I., Chen, H., Davies, F. B., D'Odorico, V., Gaikwad, P., Gallerani, S., Haehnelt, M. G., Kulkarni, G., Meyer, R. A., Welsh, L., and **Zhu, Y.**, **2026**. *E-XQR-30: evidence for an increase in the ionization state of metal absorbers from* $z \sim 6$ *to* $z \sim 2$, Monthly Notices of the Royal Astronomical Society, 546, staf2254.

78. Lin, X., Fan, X., Cai, Z., Bian, F., Liu, H., Sun, F., Ma, Y., Greene, J. E., Strauss, M. A., Green, R., Lyu, J., Champagne, J. B., Goulding, A. D., Inayoshi, K., Jin, X., Leung, G. C. K., Li, M., Liu, W., Liu, Y., Mao, J., Pudoka, M. A., Tee, W. L., Wang, B., Wang, F., Wu, Y., Yang, J., Zhang, H., and **Zhu, Y.**, **2026**. *The Discovery of Little Red Dots in the Local Universe: Signatures of Cool Gas Envelopes*, The Astrophysical Journal, 997, 364.

77. Zhang, J., Egami, E., Sun, F., Lin, X., Lyu, J., **Zhu, Y.**, Rinaldi, P., Sun, Y., Bunker, A. J., Bhatawdekar, R., Helton, J. M., Maiolino, R., Ma, Z., Robertson, B., Tacchella, S., Venturi, G., Williams, C. C., and Willott, C., **2026**. *Abundant Population of Broad Hα Emitters in the GOODS-N Field Revealed by CONGRESS, FRESCO, and JADES*, The Astrophysical Journal, 997, 250.

76. Lin, X., Egami, E., Sun, F., Zhang, H., Fan, X., Helton, J. M., Wang, F., Bunker, A. J., Cai, Z., Eisenstein, D. J., Jaffe, D. T., Ji, Z., Jin, X., Pudoka, M. A., Tacchella, S., Tee, W. L., Rinaldi, P., Robertson, B., Sun, Y., Willmer, C. N. A., Willott, C., Zhang, J., and **Zhu, Y.**, **2026**. *The Luminosity Function and Clustering of Hα Emitting Galaxies at* $z \asymp 4-6$ *from a Complete NIRCam Grism Redshift Survey*, The Astrophysical Journal, 997, 207.

75. Sun, Y., Ji, Z., Rieke, G. H., D'Eugenio, F., **Zhu, Y.**, Sun, F., Lin, X., Bunker, A. J., Lyu, J., Rinaldi, P., and Willmer, C. N. A., **2026**. *Extreme Neutral Outflow in a Non-active Galactic Nucleus Quiescent Galaxy at $z \sim 1.3$*, The Astrophysical Journal, 997, 140.

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